

Introduction

The phrase “**dynamic time warping**,” at first read, might evoke images of Marty McFly driving his DeLorean at 88 MPH in the Back to the Future series. Alas, dynamic time warping does not involve time travel; instead, it’s a technique used to dynamically compare time series data when the time indices between comparison data points do not sync up perfectly.

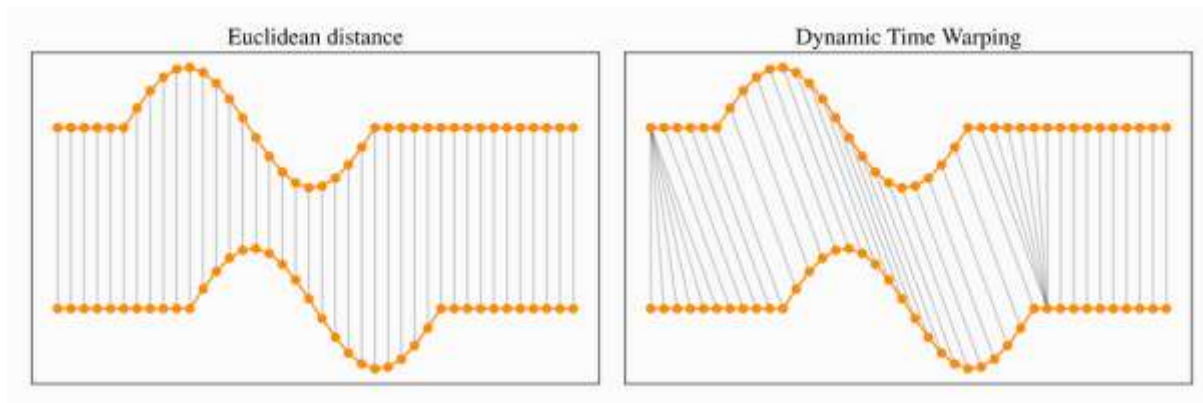
One of the most salient uses of dynamic time warping is in speech recognition – determining whether one phrase matches another, even if the phrase is spoken faster or slower than its comparison. You can imagine that this comes in handy to identify the “wake words” used to activate your Google Home or Amazon Alexa device – even if your speech is slow because you haven’t yet had your daily cup(s) of coffee.

Dynamic Time Warping (DTW)

In time series analysis, Dynamic Time Warping (DTW) is one of the algorithms for measuring the similarity between two temporal time series sequences, which may vary in speed. **The main idea of DTW is to compute the distance from the matching of similar elements between time series. It uses the dynamic programming technique to find the optimal temporal matching between elements of two-time series.**

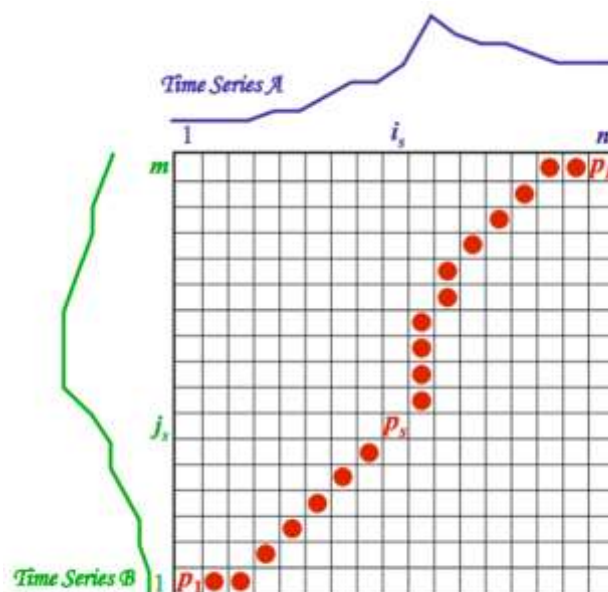
For instance, similarities in walking could be detected using DTW, **even if one person was walking faster than the other**, or if there were accelerations and decelerations during an observation. DTW has been applied to temporal sequences of video, audio and graphics data — indeed, any data that can be turned into a linear sequence can be analyzed with DTW. A well-known application has been **automatic speech recognition**, to cope with different speaking speeds. Other applications include speaker recognition and **online signature recognition**. It can also be used in partial shape matching applications.

The objective of time series comparison methods is to produce a distance metric between two input time series. The similarity or dissimilarity of two-time series is typically calculated by converting the data into vectors and calculating the Euclidean distance between those points in vector space.



In general, DTW is a method that calculates an optimal match between two given sequences (e.g. time series) with certain restrictions and rules:

1. Every index from the first sequence must be matched with one or more indices from the other sequence and vice versa
2. The first index from the first sequence must be matched with the first index from the other sequence (but it does not have to be its only match)
3. The last index from the first sequence must be matched with the last index from the other sequence (but it does not have to be its only match)
4. The mapping of the indices from the first sequence to indices from the other sequence must be monotonically increasing, and vice versa, i.e. if $j > i$ are indices from the first sequence, then there must not be two indices $l > k$ in the other sequence, such that index i is matched with index l and index j is matched with index k , and vice versa.



Time normalized distance between Series A and B

Understand DTW Mathematically

Let's assume we have two sequences like the following:

$$\begin{aligned} X &= x_1, x_2, \dots, x_i, \dots, x_n \\ Y &= y_1, y_2, \dots, y_j, \dots, y_m \end{aligned}$$

The sequences X and Y can be arranged to form an n-by-m grid, where each point (i, j) is the alignment between x_i and y_j .

A warping path W maps the elements of X and Y to minimize the *distance* between them. W is a sequence of grid points (i,j).

Warping Path and DTW distance

The Optimal path to (i_k, j_k) can be computed by:

$$D_{min}(i_k, j_k) = \min_{i_{k-1}, j_{k-1}} D_{min}(i_{k-1}, j_{k-1}) + d(i_k, j_k | i_{k-1}, j_{k-1})$$

Where d is the Euclidean distance, then the overall path cost can be calculated as:

$$D = \sum_k d(i_k, j_k)$$

Restrictions on the Warping function

The warping path is found using a dynamic programming approach to align two sequences. Going through all possible paths is “combinatorically explosive”. Therefore, for efficiency purposes, it's important to limit the number of possible warping paths, and hence the following constraints are outlined:

- **Boundary Condition:** This constraint ensures that the warping path begins with the start points of both signals and terminates with their endpoints.

$$i_1 = 1, i_k = n \quad \text{and} \quad j_1 = 1, j_k = m$$

- **Monotonicity condition:** This constraint preserves the time-order of points (not going back in time).

$$i_{t-1} \leq i_t \quad \text{and} \quad j_{t-1} \leq j_t$$

- **Continuity (step size) condition:** This constraint limits the path transitions to adjacent points in time (not jumping in time).

$$i_t - i_{t-1} \leq 1 \quad \text{and} \quad j_t - j_{t-1} \leq 1$$

In addition to the above three constraints, there are other less frequent conditions for an allowable warping path:

- **Warping window condition:** Allowable points can be restricted to fall within a given warping window of width ω (a positive integer).

$$|i_t - j_t| \leq \omega$$

- **Slope condition:** The warping path can be constrained by restricting the slope, and consequently avoiding extreme movements in one direction.

An acceptable warping path has combinations of chess king moves that are:

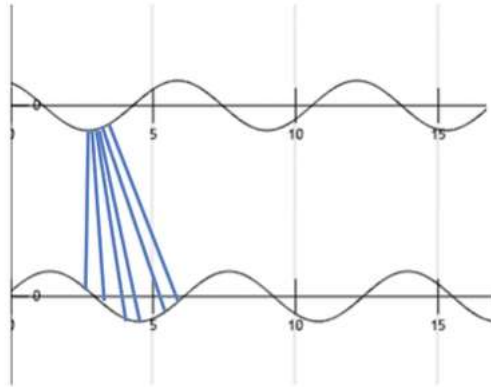
Horizontal moves: $(i, j) \rightarrow (i, j+1)$

Vertical moves: $(i, j) \rightarrow (i+1, j)$

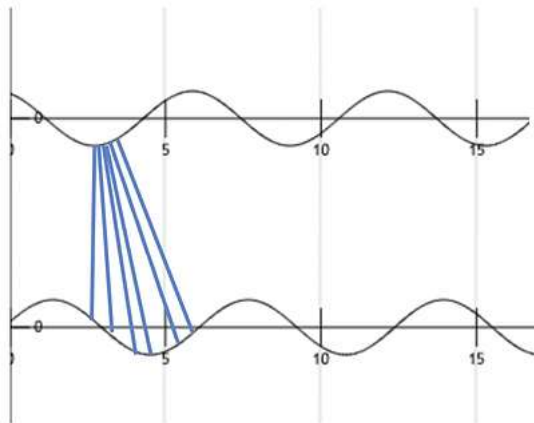
Diagonal moves: $(i, j) \rightarrow (i+1, j+1)$

Why do we need DTW?

Any two-time series can be compared using Euclidean distance or other similar distances on a one-to-one basis on the time axis. The amplitude of the first time series at time T will be compared with the amplitude of the second time series at time T. This will result in a very poor comparison and similarity score even if the two-time series are very similar in shape but out of phase in time.



DTW compares the amplitude of the first signal at time T with the amplitude of the second signal at time $T+1$ and $T-1$ or $T+2$ and $T-2$. This makes sure it does not give a low similarity score for signals with similar shapes and different phases.



Applications of DTW

1. To detect similarities in walking. If one person was walking faster than the other, or if there were accelerations and decelerations during an observation.
2. Spoken word recognition applications. They are used to match a sample voice command with others commands even if the person talks faster or slower than the pre-recorded sample voice.
3. Correlation power analysis